

EVERY ROW LEAVES A WAKE.

# WAKE - Agentic Rowing Intelligence

A visual solution dossier for rowing practitioners, hackathon judges, product reviewers, and technical evaluators.

|                                                                                     |                                                                                    |                                                                                   |                                                                                 |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <div>CLUB MEMORY</div> <div>102</div> <div>public pre/post-regatta activities</div> | <div>FIXED EVALUATION</div> <div>10</div> <div>same cases for both workflows</div> | <div>MEASURED GAIN</div> <div>+34.76</div> <div>points over direct baseline</div> | <div>SAVED REPLAY</div> <div>US\$0</div> <div>to reopen committed results</div> |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------|

WAKE

Sessions

Competition

Goal memory

Evaluation

Review a session

Overview

Attention

Team

Intelligence

Session reviews

DAILY INTELLIGENCE · CLUB WORKSPACE

Overview

Club pulse and coverage

DAILY INTELLIGENCE · TEAM AND CREW MEMORY

Understand the club, then the session.

WAKE connects athletes, recurring lineups, physical boats, and session evidence so the coach can find what needs attention without opening every chart.

MEASURED EVIDENCE

Evaluation results

RACE OUTCOME

Competition review

10 TRAINING DAYS · 17-28 AUG 2026

Club training pulse

Ten named crews share sixteen athletes across 2x, 4x, and 8x boats. WAKE surfaces the records that need attention without asking the coach to open every outing.

RECORDS SCREENED

52/52

all recorded activities

DEEP INVESTIGATIONS

2/2

verified candidate outputs

HUMAN OR SOURCE ROUTES

8

kept outside paid triage

OBSERVED AGENT COST

US\$0.19

60,094 tokens · 2 executions

Club overview: deterministic screening turns 52 activities into a coach-facing queue before deeper investigations are selected.

Evidence-backed review. Metric-level source trust. Human-approved memory.

# One file is manageable. A whole club is not.

Rowing clubs already collect valuable information, but it lives in different places: coach plans, SpeedCoach exports, Concept2 PM5 results, mobile telemetry, weather, crew lineups, boats, and conversations. The difficult task is not drawing another chart. It is reconstructing what happened, deciding which source can support each claim, exposing what remains unknown, and preserving the coach's reviewed conclusion.

WAKE is a replay-first agentic workflow for that task. It screens high-volume data deterministically, selects a bounded investigation when needed, verifies every structured output, asks the appropriate athlete or coach for missing context, and gates durable club memory behind human approval.

A coach can understand one SpeedCoach file. The value problem appears when the same coach must follow many athletes, recurring crews, physical boats, substitutions, indoor alternatives, missed sessions, weather shifts, and competition outcomes across weeks.

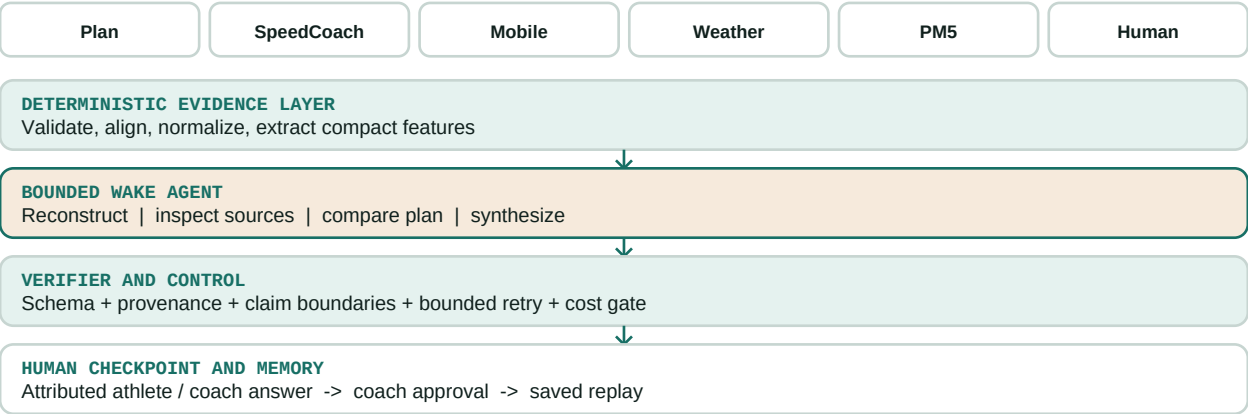
| Before WAKE                                                                          | WAKE contribution                                                                          | Human authority                                                                       |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Plans, device exports, weather, lineups, boats, and conversations remain fragmented. | Normalizes evidence, investigates contradictions, and shows only supported claims.         | Athletes provide first-hand execution context; coaches own consequential approval.    |
| A spreadsheet stores rows but does not decide what deserves attention.               | Deterministic screening selects exceptions before any paid investigation.                  | A missing session remains a question, never a fitness, injury, or commitment verdict. |
| A direct summary can sound complete even when evidence is incomplete.                | Verifier-enforced uncertainty, provenance, and source boundaries survive the final output. | Human answers add attributed context; they never rewrite measured telemetry.          |

WAKE treats coaches and athletes as complementary contributors. The athlete often owns the SpeedCoach or PM5 file and can explain equipment use or perceived effort. The coach owns the planned objective, group context, and consequential approval. Either person can upload evidence; provenance records who supplied each item.

**WAKE does not replace a qualified rowing coach. It prepares consistent, exception-focused evidence for review.**

# Bounded intelligence between evidence and memory

WAKE separates high-volume data processing from model reasoning. Source adapters validate and normalize the evidence, deterministic tools reconstruct the session and compare it with the prescription, and one bounded orchestrating agent decides which tools to call and how to synthesize the compact findings. A strict verifier rejects unsupported claims or malformed output before anything reaches the coach.



| Capability              | Implementation boundary              | Why it exists                                                    |
|-------------------------|--------------------------------------|------------------------------------------------------------------|
| Validate and normalize  | Deterministic adapters and schemas   | Reject malformed or mismatched evidence before reasoning.        |
| Reconstruct and compare | Four read-only investigation tools   | Turn high-volume rows into compact, inspectable findings.        |
| Synthesize              | One bounded orchestrating agent      | Choose tools, reconcile evidence, and draft a structured review. |
| Verify                  | Schema, provenance, and claim checks | Reject unsupported assertions and allow only a bounded retry.    |
| Approve memory          | Athlete / coach checkpoint           | Keep consequential context under accountable human control.      |

The human checkpoint is part of the architecture, not a disclaimer added at the end. Athlete and coach answers retain provenance, and durable club memory changes only after explicit coach approval. Saved verified artifacts can then be reopened and regraded without another model call.

# The best source can change by metric

Multi-sensor intelligence is not an average of every number. WAKE assigns metric-level trust. When multiple sources report SPM, the workflow compares coverage, plausibility, continuity, and consistency, then selects the best-supported source for that session and metric. GPS distance, SPM, environment, and human context may therefore come from different evidence owners.

| Evidence         | Status            | Supports                                              | Does not prove                                    |
|------------------|-------------------|-------------------------------------------------------|---------------------------------------------------|
| Training plan    | Core              | Intent, intervals, target SPM, recovery, objective    | What was actually completed                       |
| SpeedCoach       | Core              | Dedicated rowing pace, distance, stroke-rate evidence | Crew context or visible technique                 |
| Mobile telemetry | Optional enhancer | Route, timing overlap, corroborating sensors          | Automatic authority over a more consistent source |
| Environment      | Optional enhancer | Time-aligned wind, gust, temperature, humidity        | That weather caused a performance change          |
| Concept2 PM5     | Athlete-owned     | Indoor pace, watts, SPM, workout structure            | On-water speed, technique, strength, or health    |
| Human context    | Attributed        | Equipment, effort, interruption, crew observation     | A rewrite of device measurements                  |

## Every statement is typed

| Observed        | Derived                 | Human context     | Hypothesis       | Action                  | Unknown      |
|-----------------|-------------------------|-------------------|------------------|-------------------------|--------------|
| Direct evidence | Transparent calculation | Attributed answer | Marked inference | Coach-reviewable action | Explicit gap |

High-volume sensor rows are processed deterministically. Raw telemetry dumps are not sent directly to the model.

# Start at club scale

The product story moves from club scale to a single investigation and back to longitudinal memory. The coach first sees coverage, exceptions, crews, and athletes. A selected session then exposes evidence ownership, reconstructed intervals, metric-level source authority, environmental limits, and the smallest unresolved human question.

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Club overview: deterministic screening turns 52 activities into a coach-facing queue before deeper investigations are selected.

PRE - REGATTA

52

validated and reconstructed activities

POST - REGATTA

50

additional longitudinal activities

ATHLETES

16

fictional identities in the public demo

CREWS

10

recurring 2x, 4x, and 8x lineups

# People, lineups, boats, and training days

The public demonstration connects sixteen fictional athletes, ten recurring crews, eleven physical boats, and training across 2x, 4x, and 8x classes. Names and exact outcomes are fictional; formats and plausible patterns were informed by supplied coach plans, SpeedCoach files, PM5 photographs, and first-hand club context.

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Team

Crews, affiliates, and boats

ATHLETE-CENTERED CHRONOLOGY

Training days connect water and indoor work.

Volumes remain separated by modality.

POB CONTEXT

ACTIVE ATHLETE-DAYS

108

Individual days with a recorded activity

COMBINED DAYS

7

water plus Concept2 on the same plan-confirmed day

INDOOR-ONLY DAYS

7

valid training, not a missing water session

MODALITY VOLUME

489.5 / 106.1 km

water / indoor - never merged

TEAM AND CREW HERST

Every outing stays attached to people and a physical boat.

Select a crew to inspect its lineup and two-week history.

2x

4 CREWS

RECURRING LINEUP

Crew: Tucano - 2x - Men

Launched

Lucas - Rafael

3/4

LAUNCHED

1 TO REVIEW

RECURRING LINEUP

Crew: Arara - 2x - Women

Marina - Helena

4/4

LAUNCHED

NO MATERIAL PLANS

Team memory connects recurring crews, athletes, physical boats, and indoor training without merging unlike modalities.

WAKE

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Back

CURRENT LOCATION

WAKE / Athlete / Lucas

HEN SQUAD - ATHLETE MEMORY

Lucas

WAKE connects this athlete's water, indoor, crew, and physical-boat history while keeping each modality's evidence and meaning separate.

SYNTHETIC MEM

ACTIVE DAYS

8/10

2 combined - 0 indoor-only

WATER DISTANCE

130.0 km

10 water sessions

INDOOR DISTANCE

13.0 km

2 individual Concept2 records

EXPECTED GAPS

0

content requested, never treated as a fitness verdict

CREW REPERSECTS

3 recurring lineups

Crew: Tucano - 2x - Men

Mon - 2x

Open crew

Crew: Gaviao - 4x - Men

Mon - 4x

Open crew

Crew: Carcara - 8x - Men

Mon - 8x

Open crew

PHYSICAL BOATS BOWED

Gavião - Tucano - Biguá - Carcará

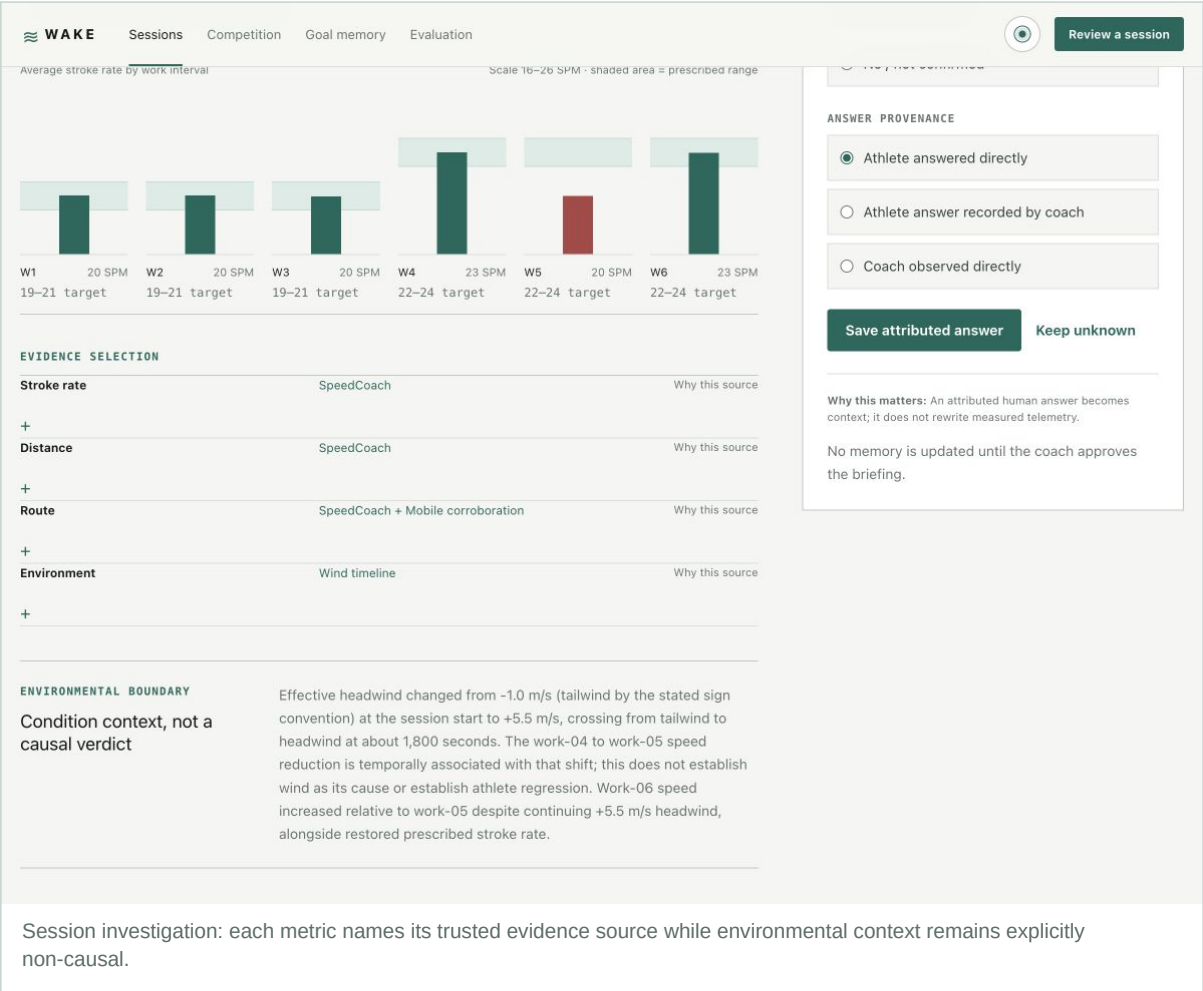
Boat names come from linked session content. A class such as 2x is not treated as the identity of the physical shell.

Athlete history preserves crew, solo, and Concept2 work in one chronology while keeping water and indoor distance separate.

| Relationship retained                            | Product value                                                                   |
|--------------------------------------------------|---------------------------------------------------------------------------------|
| Athlete -> crew -> physical boat -> outing       | A coach can inspect shared history without losing individual ownership.         |
| Athlete -> solo water / Concept2 -> training day | Indoor-only work is valid training, not a missing-water-session alert.          |
| Crew snapshot -> race entry -> official field    | A later result retains who raced together without rewriting training causality. |

Concept2 results belong to one athlete even when a prescription is shared. Equivalent PM5 workouts can support within-workout comparisons of pace, SPM, and watts. They do not directly measure on-water speed, visible technique, muscular strength, injury, or medical fitness.

# Reconstruct, verify, and ask the right person



ANSWER PROVENANCE

☒ Athlete answered directly

☐ Athlete answer recorded by coach

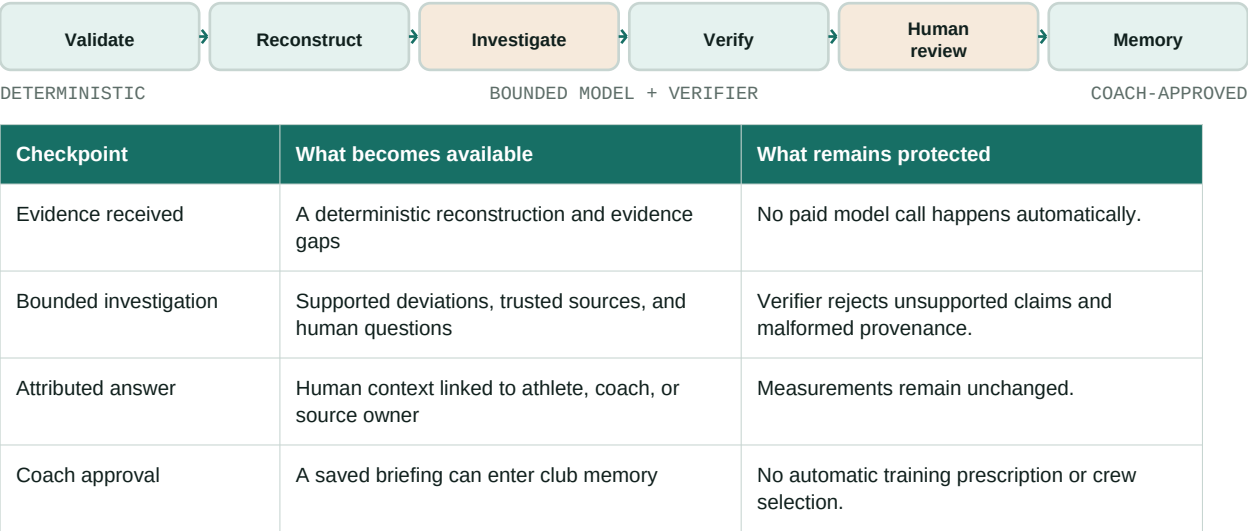
☐ Coach observed directly

Save attributed answer

Keep unknown

Why this matters: An attributed human answer becomes context; it does not rewrite measured telemetry.

No memory is updated until the coach approves the briefing.



# Close the loop without inventing causation

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GOAL READINESS → RACE OUTCOME → NEXT CYCLE

# Competition Review

Connect every club entry to its athletes, boat, previous shared work, official result, and the full competitive field.

OUR ENTRIES

10

8 events

ATHLETES

16

16 with multiple starts

WINS

3

8 podium finishes

CLASSIFIED

9/10

1 needs context

OPPOSITION

3

fictional competitor clubs

OUR ENTRIES

Ten boats, one connected club history

Open any boat to inspect the field and its pre-race evidence.

RACE 1 · 2X MEN BEGINNER

2

Crew: Tucano - 2x - Men

OF 4

Lucas · Rafael

3 shared outings · 35.0 km contextual history

1:49.8

1:49.8 / 500 m

Open boat report →

RACE 1 · 2X WOMEN BEGINNER

1

Crew: Arara - 2x - Women

OF 4

Marina · Helena

4 shared outings · 47.5 km contextual history

1:52.5

1:52.5 / 500 m

Open boat report →

RACE 4 · 2X MIXED ASPIRANT

2

Crew: Bem-te-vi - 2x - Mixed

OF 5

Bruno · Camila

2:06.3

2:06.3 / 500 m

Open boat report →

COMPETITIVE FIELD

Eight events reconstructed

Results retain opponent club, crew, athlete identities, official order, time, and non-completion. Raw time never replaces the published rank.

R1 · 2x Men Beginner

500 m · 4 entries

1:46.4

R1 · 2x Women Beginner

500 m · 4 entries

1:52.5

R4 · 2x Mixed Aspirant

1,000 m · 5 entries

4:08.1

R12 · 4x Men Master D

1,000 m · 5 entries

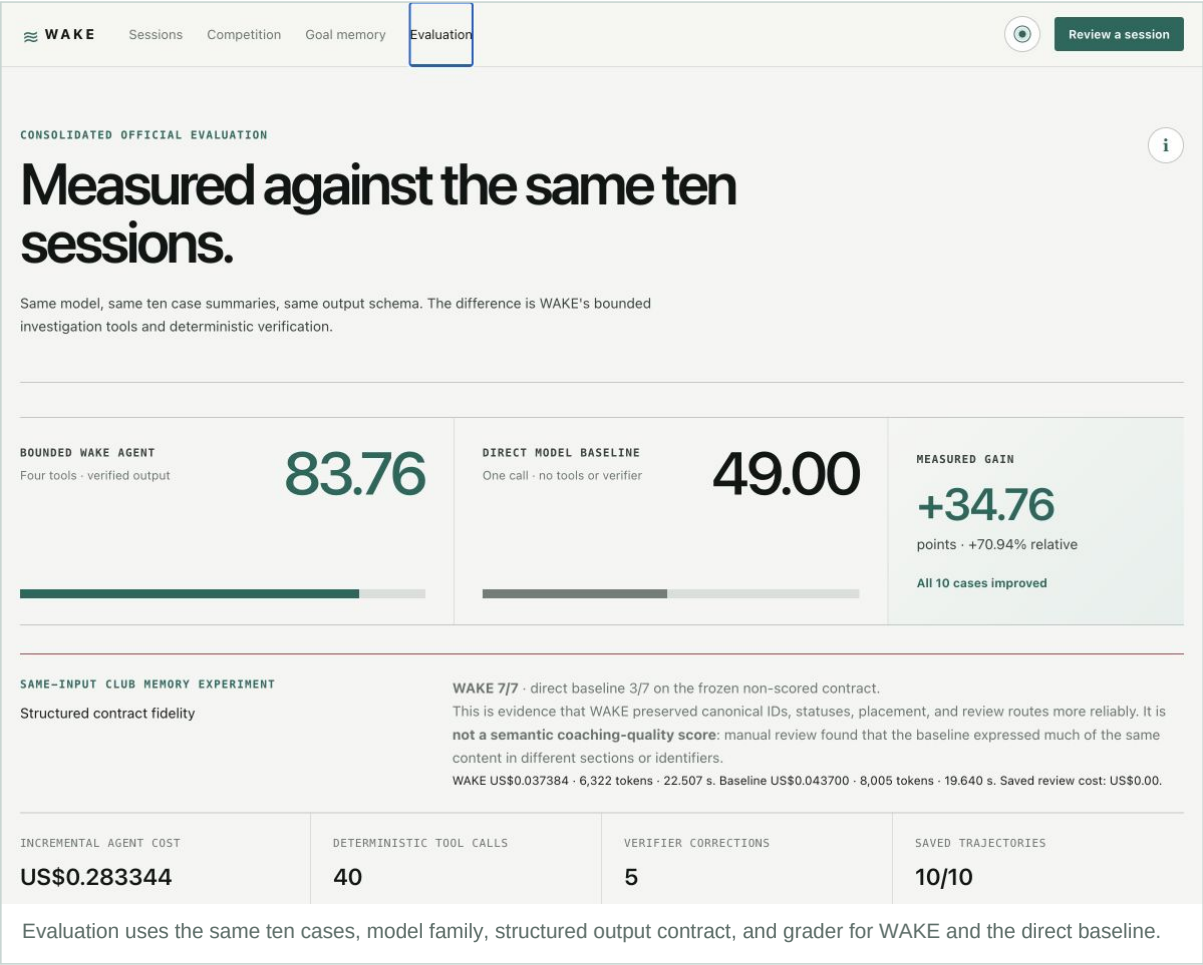
3:56.8

Competition review closes the training loop with crew snapshots, boats, opponents, official order, and honest context boundaries.

Ten fictional club entries across eight fictional events connect all sixteen athletes to exact lineup snapshots, physical boats, pre-race shared outings, opponent clubs, official order, pace, gaps, and one context-seeking non-completion. Category distances are informed by supplied competition material, while identities and displayed outcomes remain fictional.

| WAKE can show                                                                                               | WAKE must not claim                                                                  |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Training context before the race, the exact lineup, boat, opposition, official order, and missing evidence. | That one session, crew change, or weather condition caused the result.               |
| Comparable observations across periods when the evidence contract matches.                                  | Automatic selection, athletic potential, injury, physiology, or medical conclusions. |

# Measured against the same ten sessions



| Workflow        | Score         | Cases            | Observed cost    | Interpretation                                                      |
|-----------------|---------------|------------------|------------------|---------------------------------------------------------------------|
| Direct baseline | 49.00 / 100   | 10               | Reference        | One structured model call; no investigation tools or verifier.      |
| Bounded WAKE    | 83.76 / 100   | 10               | +US\$0.283344    | Four tools, verification, claim boundaries, and saved trajectories. |
| Measured gain   | +34.76 points | 10 / 10 improved | +70.94% relative | Workflow evidence only; not a human-coach comparison.               |

The result is not uniformly positive. Environmental interpretation regressed from 80.00 to 76.00, and the single real anonymized case remains WAKE's weakest at 53.71. These results are preserved rather than edited away after inspection.

## Failures remained part of the evidence

The project records experiments that failed, regressed, or changed direction. This is part of the evidence, not an appendix of excuses. Examples include an environmental-scoring regression, a longitudinal pilot with no demonstrated quality gain, early navigation that was not bookmarkable, oversized explanatory banners, and an initial assumption that a few cases could describe the solution.

| Experiment or failure             | Observed evidence                                                                                      | Decision kept in the product                                                                                     |
|-----------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Three-case framing                | Too narrow to express wind, changing conditions, incorrect execution, crews, and longitudinal context. | Expanded to ten fixed evaluations plus a 102-activity relational club demonstration.                             |
| Core reconstructed-distance claim | The first evidence ablation overinterpreted derived segment margins as completed-distance shortfall.   | Added a failing regression test and verifier-enforced distance boundary; v2 passed 8/8, 10/10, and 12/12 checks. |
| Longitudinal reasoning pilot      | WAKE and baseline both passed the same non-scored capability checks.                                   | Preserved<br>NO_DEMONSTRATED_QUALITY_GAIN;<br>report only the observed 29.01% cost reduction.                    |
| Environment interpretation        | Official dimension score regressed from 80% to 76%.                                                    | Keep weather associative, visible, and explicitly non-causal.                                                    |
| More autonomous agents            | No submitted evidence showed specialization improved the MVP.                                          | Keep one bounded orchestrator until evaluation justifies extra complexity and cost.                              |

| RED                                            | GREEN                                        | REFACTOR                                              | EVALUATE                                                            |
|------------------------------------------------|----------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------|
| Write a failing behavioral or regression test. | Implement the smallest deterministic change. | Preserve boundaries while the full suite stays green. | Run fixed cases and record gains, regressions, cost, and decisions. |

**A failed experiment can still improve the product when its expected behavior, evidence, cost, and decision are preserved.**

# A judge can reopen the evidence without a key

WAKE spends model budget only after deterministic screening identifies a useful investigation. Every live start requires explicit authorization and shows the observed cost afterward. Saved outputs are application-owned artifacts and can be reopened or regraded offline without another model call.

| Artifact or run                               | Observed model cost         | Can reopen without a call?                 |
|-----------------------------------------------|-----------------------------|--------------------------------------------|
| Official ten-case incremental WAKE comparison | US\$0.283344                | Yes                                        |
| Saved 102-activity club memory                | US\$0.037384 / 6,322 tokens | Yes, US\$0.00                              |
| Two saved demo-club investigations            | US\$0.194118 combined       | Yes, US\$0.00                              |
| Clean reproduction and dashboard replay       | US\$0.00                    | Yes; no OPENAI_API_KEY required            |
| Optional new live investigation               | Authorized per start        | The new output is saved after verification |

## Clean judge path

```
./scripts/reproduce_submission.sh
./scripts/start_dashboard.sh
# open http://localhost:3000/
# replay and saved evaluation cost US$0.00
```

WAKE does not replace a qualified rowing coach. It does not autonomously select crews, diagnose injury or physiology, infer commitment from absence, or evaluate visible technique without appropriate video or biomechanical evidence.

| Public submission includes                                                                                         | Public submission excludes                                                                                  |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Synthetic fixtures, minimized approved references, saved outputs, trajectories, tests, schemas, prompts, and PDFs. | API keys, .env, private GPS, device identifiers, identifiable health data, and raw private athlete records. |

The reproducible MVP demonstrates the value chain from fragmented evidence to verified session review, club memory, longitudinal comparison, competition context, and fixed-case evaluation. The final interface narrative and five-minute video have completed owner QA.

Every row leaves a wake. WAKE makes that trail inspectable without pretending uncertainty has disappeared.